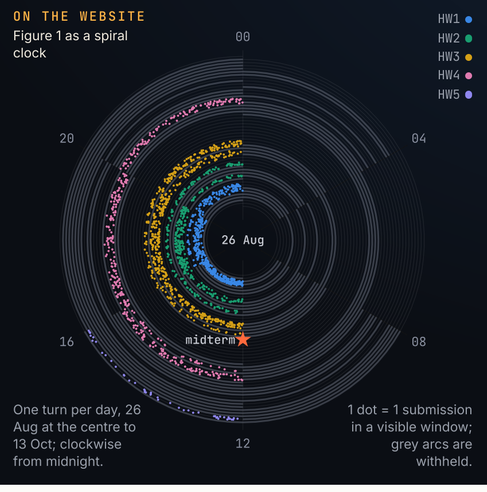
STATS 401 · FINAL PROJECT · CS201 HOMEWORK PROCESS

The score *has a past*

Homework scores cluster near full marks. We followed **3,359** online-judge submissions to see how differently students get there, and whether the path travels with the exam.

Zaozao Wang data acquisition, cleaning, analysis **Zhengxiang Liu** visualization design, D3 implementation

ON THE WEBSITE
Figure 1 as a spiral clock



One turn per day, 26 Aug at the centre to 13 Oct; clockwise from midnight.

1 dot = 1 submission in a visible window; grey arcs are withheld.

Watch it move
The playable data film, all twelve figures, tables and methods

WHY IT MATTERS

In 36 of 43 problem slots, every student who attempted the problem reached full marks in the end. What the 100 does not show is the first try, the retries, or the hour they arrived.

The online judge keeps that process in its submission log. Reading it helps students plan their own work, and helps TAs and instructors see where and when support is needed.

QUESTIONS

- Q1 When does homework work happen, and is a busy hour many students or a few students submitting again and again? — Fig 1
- Q2 Do retries pay off, and does that depend on the homework, the time since the last try, or the point in the window? — Fig 2 · 3
- Q3 What does a final 100 hide about how students got there? — Fig 4
- Q4 Does the way students submit relate to their midterm score? — Fig 5

DATA & METHOD

SOURCE CS201 online judge, read-only snapshot of 13 Oct 2026: 3,622 course submissions from a 42-student roster; **3,359** inside five homework windows (39 submitters, 43 problem slots).

CLEANING Test account removed; each submission kept only inside its own homework window; scores scaled to 0–100 per problem; judge errors skipped.

ANALYSIS First try, best within three, best observed; 1,345 eligible retries; four-hour windows; five phases per homework. Exam: 26 grades linked privately; Spearman ρ, 5,000-draw bootstrap.

PRIVACY Only aggregates are published; any group of fewer than five students is withheld.

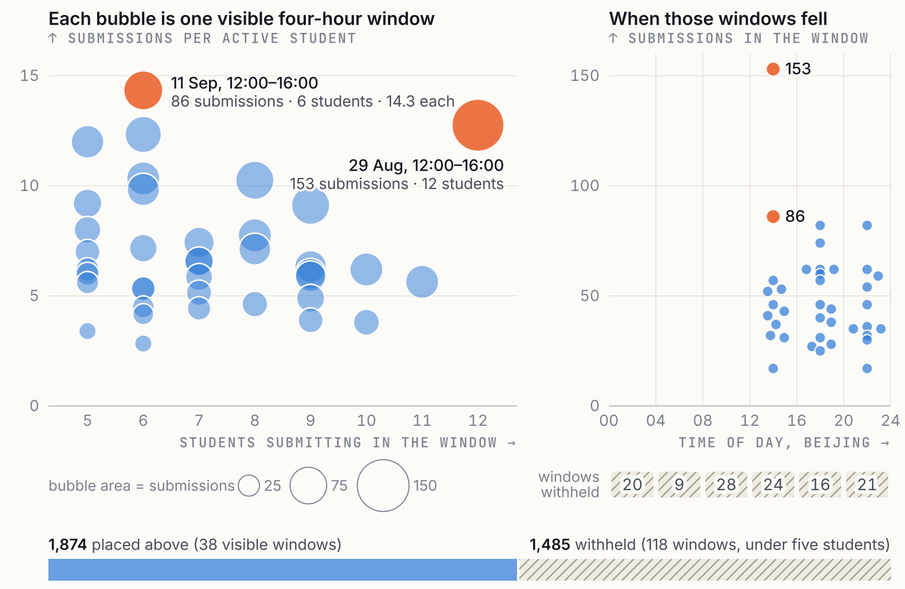
BUILT WITH Python for cleaning; D3 v7, SVG and Canvas for the film and this poster. Online: switch views, sort, highlight a homework, follow a problem across figures, hover for counts, scrub the story.

PROCESS how submissions and retries unfold

1 WHEN THE WORK ARRIVES

Film · Figure 01

A burst can be many students, or a few students many times.



HOW TO READ One bubble is one four-hour window, such as 29 Aug, 12:00–16:00, in which at least five students submitted. Further right: more students. Higher: more submissions per student. Bigger: more submissions in total.

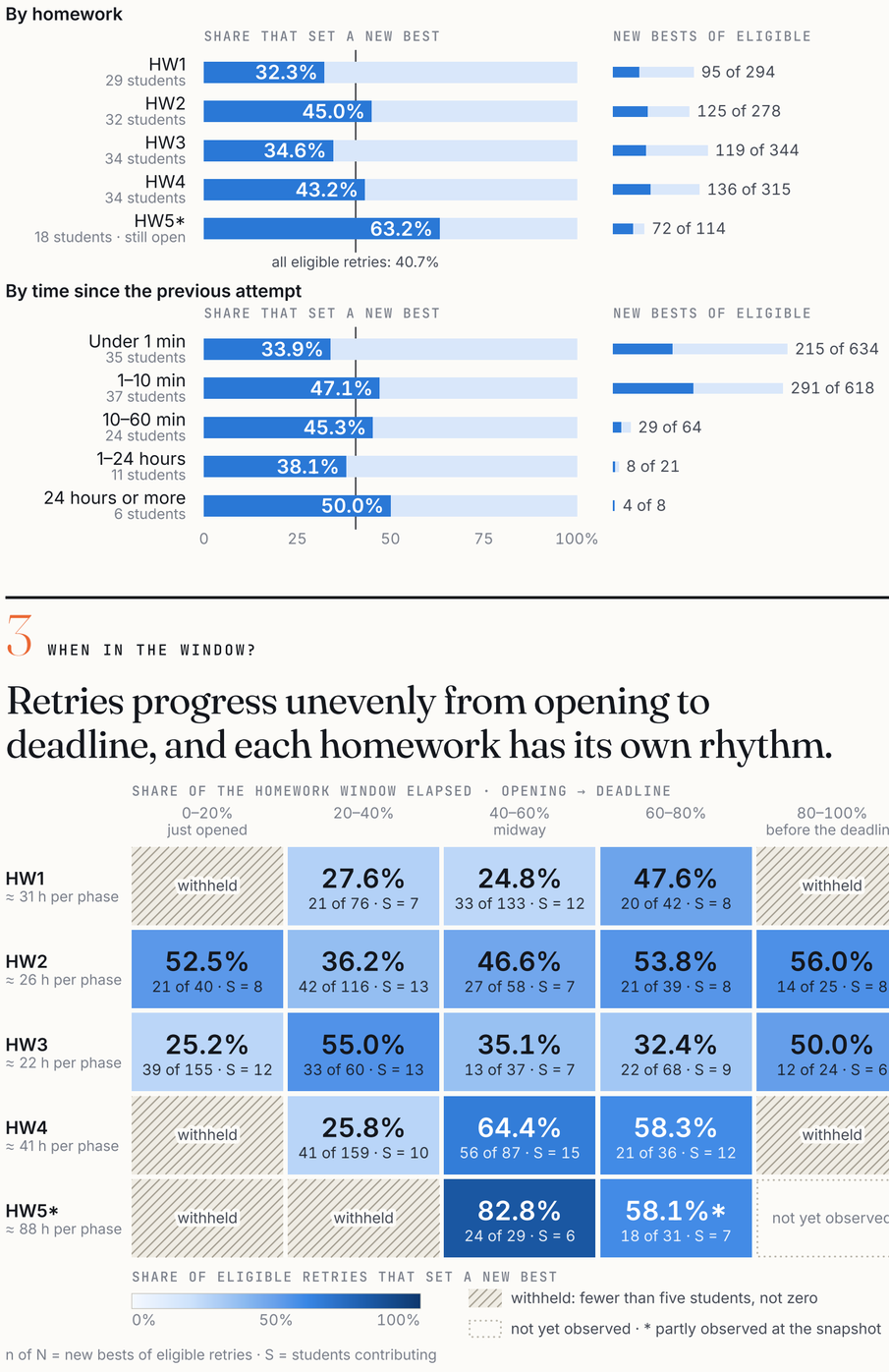
WHAT WE SEE The largest visible window holds **153** submissions from **12** students; the most intense, **86** from only **6**, or **14.3** each. Every published window falls between noon and midnight.

CAREFUL Windows with fewer than five students are withheld for privacy: **1,485** submissions in 118 windows, 57 of them before noon. Counted, never placed, never zero. Clock time is not study time.

2 DO RETRIES PAY OFF?

Film · Figure 11

547 of 1,345 eligible retries set a new personal best: 40.7%.



HOW TO READ An eligible retry is a scored resubmission made while the student's best on that problem is still below 100. The dark bar is the share that beat every earlier score; on the right are the counts behind each share.

WHAT WE SEE **47%** of these retries came less than a minute after the previous attempt, and they set a new best least often: **33.9%**, against **47.1%** after 1–10 minutes.

CAREFUL Retries are repeated events from the same students, not independent trials. Gaps are clock time, not study time, so these rates do not estimate an effect of waiting. Long gaps have few retries: read the counts. * HW5 is still open.

3 WHEN IN THE WINDOW?

Film · Figure 12

Retries progress unevenly from opening to deadline, and each homework has its own rhythm.

HOW TO READ Each row is a homework. Its open period, from release to deadline, is cut into five equal parts: 0–20% is the first fifth, 80–100% the last. A cell is the share of that part's eligible retries that set a new best (darker is higher), with the counts and *S*, the number of students.

WHAT WE SEE Visible rates run from **24.8%** to **82.8%**, and no single pattern repeats across the homeworks.

CAREFUL Hatched cells have fewer than five students: withheld, not zero. A phase lasts 22–88 hours depending on the homework. Deadlines are the configured ones; individual extensions are unknown. No deadline effect is estimated.

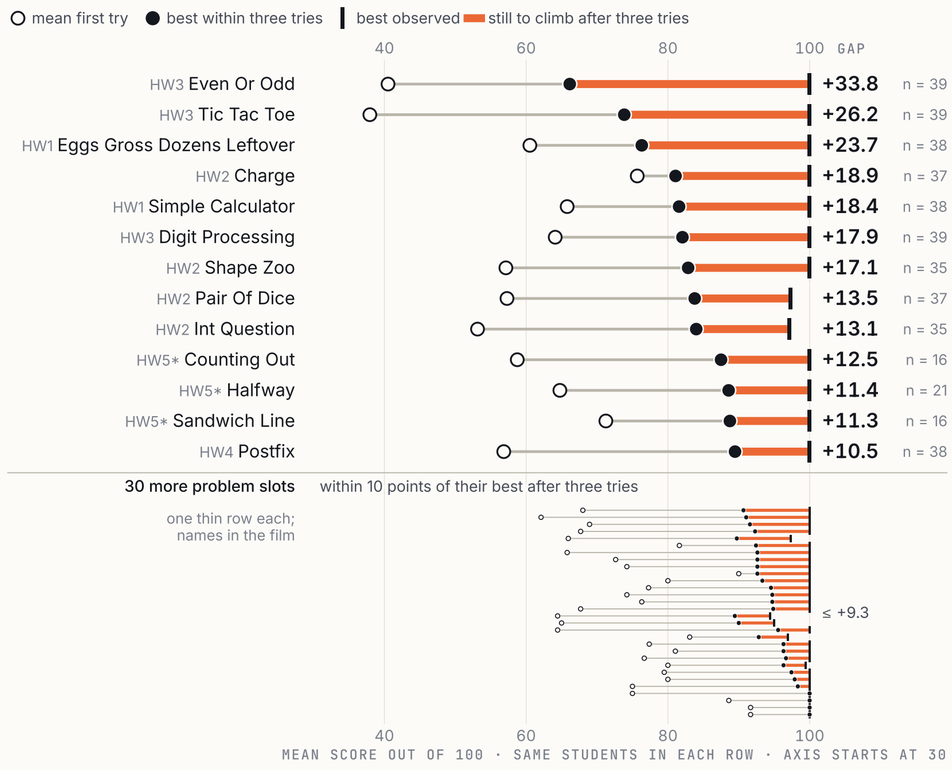
OUTCOME score gaps and the midterm

4 WHAT THE 100 HIDES

Film · Figure 03

After three tries, 13 of 43 problems still trail their best score by more than 10 points.

HOW TO READ One row per problem, averaged over the same students. Orange is what was still missing after three tries.



WHAT WE SEE The largest remaining gap is **33.8** points, for Even Or Odd (HW3). The lowest first try, **37.9**, was on Tic Tac Toe (HW3).

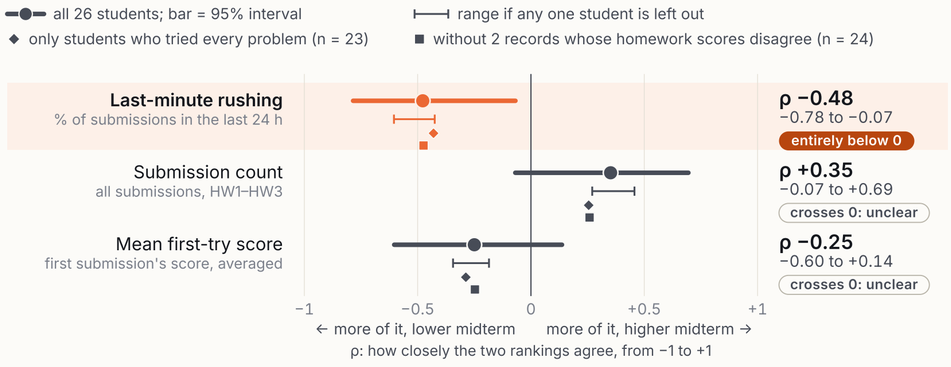
CAREFUL "Best within three" is a cumulative best. It is not the score of the third submission, and not a causal gain. * HW5 is still open: its rows are the scores observed so far.

5 DOES RUSHING SHOW IN THE EXAM?

Film · Figure 06

Last-minute rushing went with lower midterm scores.

HOW TO READ Each row is one correlation across the same 26 students, each with one value for the behaviour and one midterm score; no one is split into groups. ρ (rho) compares the two rankings: −1 opposite order, 0 no relation, +1 same order. A bar that crosses 0 leaves the direction unclear.



WHAT WE SEE Rushing, the share of a student's submissions sent in the last 24 hours before each deadline, has ρ = **−0.48** (95% interval −0.78 to −0.07): the only bar entirely below 0. Leaving out any one student keeps it between −0.60 and −0.42.

CAREFUL "Rushing" names submission timing, not effort or study time. Grades were supplied for 26 of 41 eligible students: a partial, non-random sample, using pre-exam HW1–HW3 only. An association in a small group, not a cause and not a prediction.

READ WITH CARE

- Observational data from one course: associations, not causes or predictions.
- Submissions are not effort; time between them is not study time.
- Groups of fewer than five students are withheld, never shown as zero.
- HW5 is still open until 18 Oct: its numbers are observations so far.

What the *100* was hiding

One score, many roads.
FIGURES 1 · 4
Full marks hide very different paths: busy windows and a few students submitting again and again; problems solved at once and problems still short after three tries.

Retries progress unevenly.
FIGURES 2 · 3
40.7% of eligible retries set a new best, at rates that differ by homework, by time since the last attempt and by phase of the window.

Rushing went with lower scores.
FIGURE 5
Among 26 supplied grades, more last-minute submitting went with lower midterm scores (ρ −0.48): an association, not a cause.